Sensitive Attribute Reconstruction from a Rashomon Set of Decision Trees

1 Method

1.1 Rashomon set construction

We binarize a curated subset of the UCI Adult dataset: the sensitive target T = `sex` (single indicator column), background-knowledge attribute A = `marital-status` (collapsed to 4 categories: Married, Never-married, Divorced, Widowed-Separated), background-knowledge attribute B = `education` (collapsed to 5 categories: Below-HS, HS-grad, Some-college, Bachelors, Grad+), plus `age` and `hours-per-week` median-split indicators and a `capital-gain`> 0 indicator – 13 binary columns in total. The prediction label Y is income (> \$50K vs. not).

This curated feature set is a deliberate scope reduction from the full ∼60-column binarization of Adult: with all columns included, TreeFARMS (Xin et al., 2022) only produced trees shallow enough (1–2 splits) at laptop-tractable regularization that none of them ever split on `sex`, making every background-knowledge condition identical by construction. Reducing the competing feature count lets T, A, and B actually get selected as splits while keeping the exhaustive Rashomon-set search tractable.

For each of 200 repetitions (seeds), we sample $n = 100$ rows from Adult and build a Rashomon set with TreeFARMS using regularization $\lambda = 0.01$ (the same value Hsu et al., 2026, use for Adult) and Rashomon bound multiplier 0.1 (all trees within 10% of the optimal objective are kept). This produces substantially deeper, more expressive trees than an earlier pass of this experiment that used $\lambda = 0.02$, bound multiplier 0.2: average leaf depth rose from 1–3 splits to 5.7 splits, and in a 200-tree sample, 100% of trees split on `sex` at least once (vs. 30% before). Since TreeFARMS's own search is single-threaded, the 200 repetitions are instead parallelized across seeds (one process per seed, 12-way) to fit within a practical wall-clock budget (200 repetitions complete in about 45 minutes); each process pins its CP-SAT solves to a single thread to avoid oversubscribing the machine. For every (seed, m) cell we also log the fraction of released trees containing a leaf whose root-to-leaf path splits on both `sex` and A (resp. B), used in Section 2 to test a hypothesized mechanism directly.

1.2 Reconstruction attack

We reimplement the constraint-programming reconstruction attack of Ferry et al. (2024, *"Trained Random Forests Completely Reveal your Dataset"*, known as DRAFT), adapted to read TreeFARMS's native tree representation and to support fixing a subset of feature columns to their true value for every row (the attacker's background knowledge). The attack is solved with OR-Tools CP-SAT (no commercial solver license required): one boolean variable per (row, feature), one-hot-sum constraints per categorical attribute, and per-tree, per-leaf constraints forcing the solved table to reproduce the exact per-class training-example counts of the released trees. The label Y is assumed known to the attacker in every condition, consistent with this literature: the released trees predict Y, so Y is treated as public.

For a released ensemble of the first m trees from the Rashomon set (TreeFARMS's enumeration order; no diversity-aware selection strategy is applied), we solve the attack under four background-knowledge conditions, holding the data sample and released trees fixed within a repetition:

- **None**: neither A nor B is known.
- **A**: A is known for every row; B is not.
- **B**: B is known for every row; A is not.
- **A+B**: both A and B are known.

In every condition T remains unknown. We sweep $m \in \{5, 10, 25, 50, 100\}$, with a 120-second solver timeout per (seed, m, condition) cell.

1.3 Evaluation

The solver has no notion of row identity beyond Y (fixed per row index) and whichever columns are pinned as known; under partial background knowledge, rows sharing a label are otherwise interchangeable as far as the model's constraints are concerned. Before scoring T, we therefore match each reconstructed row to a true row via the Hungarian algorithm (within each Y-label group), using Hamming distance over every column *except* T as the matching cost – excluding T from the matching cost avoids letting the evaluator's own knowledge of the true values inflate T's measured accuracy. Reconstruction error is the fraction of matched rows whose reconstructed T disagrees with the truth.

We compare against two baselines: a uniform-random guess, and a holdout-marginal guess that samples T from its empirical marginal estimated on a 2000-row holdout sample disjoint from the attacked 100 rows and never passed to the Rashomon-set construction or the attack.

1.4 Per-row confidence

The CP-SAT solver returns *a* table consistent with the released leaf counts, not a calibrated distribution over compatible tables the way Ferry et al. (2024a)'s entropy-based method does for a single tree – no reviewed paper measures this for an ensemble of released models. We approximate a confidence signal by re-solving the same (seed, m, condition) cell 15 times with different solver seeds (different search order, so a different feasible solution whenever more than one exists) and, for each true row, row-matching every repeat's output independently and recording how often the matched reconstruction of T agrees with its own majority vote – this agreement rate is the row's confidence score, and we check whether the majority-vote value is actually correct. Matching each repeat independently (rather than once across repeats) is necessary because each solve's output has its own, unrelated row ordering (Section 1.3). We run this for 20 seeds at $m \in \{10, 100\}$ (sparse vs. rich release), all four conditions, pooling rows across conditions and seeds for the calibration analysis.

The raw agreement rate is not itself a calibrated probability (Section 2.3 shows it systematically overstates accuracy). We therefore split the 20 seeds into 14 training and 6 held-out test seeds, fit an isotonic regression (agreement rate $\to P(\text{correct})$, separately per m) on the training seeds only, and evaluate the resulting calibrated probabilities – against the raw agreement rate as a baseline – purely on the held-out test seeds, using Brier score and expected calibration error (ECE, 10 bins).

2 Results

2.1 Background knowledge of A and/or B significantly reduces reconstruction error on T

Figure 1 shows the mean reconstruction error on T across 200 repetitions, for each background-knowledge condition and number of released trees m, against the two baselines (means taken over all repetitions, since the baselines do not depend on m).

The released trees leak substantially more than either baseline. All four conditions solve to error between 0.31 and 0.36 across every m tested, against a uniform-random baseline of 0.50 and a holdout-marginal baseline of 0.44.

A clean ordering: $A{+}B < A \approx B <$ None. The four conditions separate cleanly and consistently from $m = 10$ onward (Table 1), with $A{+}B$ giving the lowest error at every m tested.

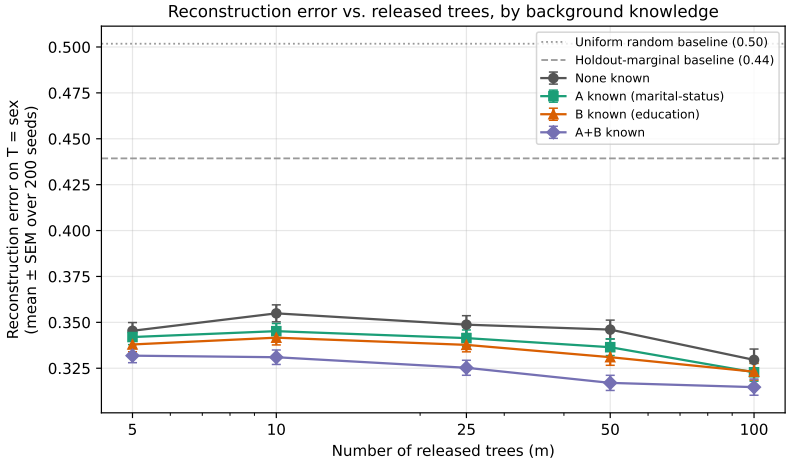


Figure 1: Reconstruction error on $T = \texttt{sex}$ vs. number of released trees m, by background-knowledge condition. Error bars are ± 1 standard error of the mean over 200 repetitions (fewer at $m = 50, 100$ where some repetitions' Rashomon sets or solves did not reach that point – see Limitations).

	$m = 5$	$m = 10$	$m = 25$	$m = 50$	$m = 100$
None	0.345	0.355	0.349	0.346	0.330
A	0.342	0.345	0.341	0.336	0.323
B	0.338	0.342	0.338	0.331	0.323
$A{+}B$	0.332	0.331	0.325	0.317	0.315

Table 1: Mean reconstruction error on T by condition and m, over 200 repetitions.

A paired Wilcoxon signed-rank test (same seed and m, condition vs. *None*), pooled across all m, confirms this is not noise:

	A	B	$A{+}B$
mean difference vs. None	-0.0077	-0.0110	-0.0211
p-value (pooled)	6×10^{-4}	3×10^{-7}	1×10^{-22}

Because this pools 15 tests (3 conditions $\times 5$ values of m), we additionally report per-m tests with Holm-Bonferroni correction for multiple comparisons ($\alpha = 0.05$): $A{+}B$ survives correction at $m = 5, 10, 25, 50$ (4 of 5 values; at $m = 100$ it narrowly fails, $p = 0.0097$, just short of its step-down threshold), and B survives at $m = 10$. A alone does not survive the conservative per-m correction at any single m, despite being significant pooled – its effect size per cell ($\approx$ 0.008–0.01) is simply too small relative to per-cell noise

($n = 200$ per cell) to clear a corrected threshold alone, even though it reliably accumulates to significance when pooled. We read the overall pattern as: knowing *both* A and B gives a robust, individually-significant effect; knowing either one alone gives a real but smaller effect that is only reliably detectable in aggregate. B (education) is directionally the noisier of the two single-attribute conditions, consistent with a weaker empirical association with `sex` than marital-status has in this data.

2.2 Testing the hypothesized mechanism: it does not hold up

An earlier pass of this experiment (fewer, shallower trees) found no condition separation, and we hypothesized that background knowledge only helps where a released tree's root-to-leaf path jointly splits on T *and* A or B together – deeper trees seemed to confirm this indirectly, since they both showed much more joint splitting and the first clear condition separation.

We tested this directly rather than leaving it as a plausible story. For every (seed, m) cell we computed the fraction of released trees containing at least one leaf whose path uses both T and the attribute in question, and correlated it against that cell's observed error reduction (*None* error minus the condition's error). Figure 2 shows the result: no relationship. Both attributes show a flat scatter (Spearman $\rho = 0.03$, $p = 0.42$, $n = 949$ for A; $\rho = 0.04$, $p = 0.22$, $n = 950$ for B).

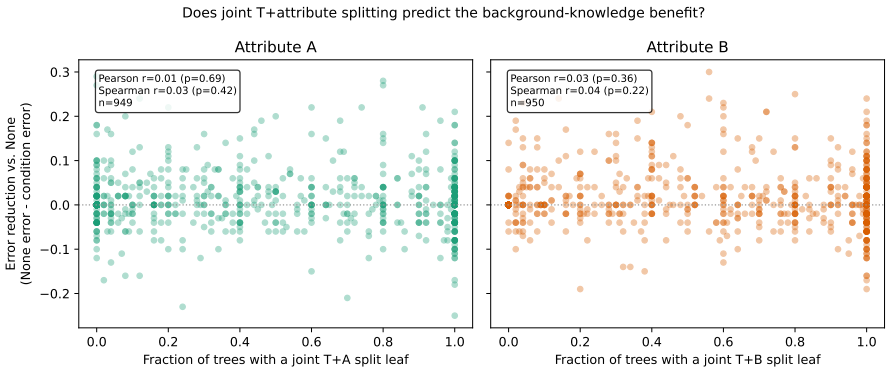


Figure 2: Per-(seed, m) error reduction vs. fraction of released trees with a joint T+attribute split leaf. No correlation is visible for either attribute.

We take this as a genuine negative result, not a nuisance to explain away: the simple structural story ("more joint splits ⇒ more leakage") does not predict cell-to-cell variation, even though the aggregate effect across hundreds of cells is unambiguous (previous subsection). The most likely reading is that single-cell reconstruction error is dominated by solver and sample idiosyncrasies that swamp this particular structural signal, so the

background-knowledge benefit is real on average but not well explained by this one coarse proxy – a deeper mechanism account (e.g. weighting joint-split leaves by how many training rows they actually capture, rather than a binary per-tree indicator) is left for future work.

2.3 Per-row confidence: solver agreement predicts correctness

No reviewed paper reports a confidence value for an individual reconstructed record from an ensemble of released models (gap G1 identified in the literature review); the solver-agreement proxy described in Section 1.4 is our attempt to fill that gap. Figure 3 bins the 20-seed, $m \in \{10, 100\}$ sample by agreement rate and plots each bin's empirical accuracy.

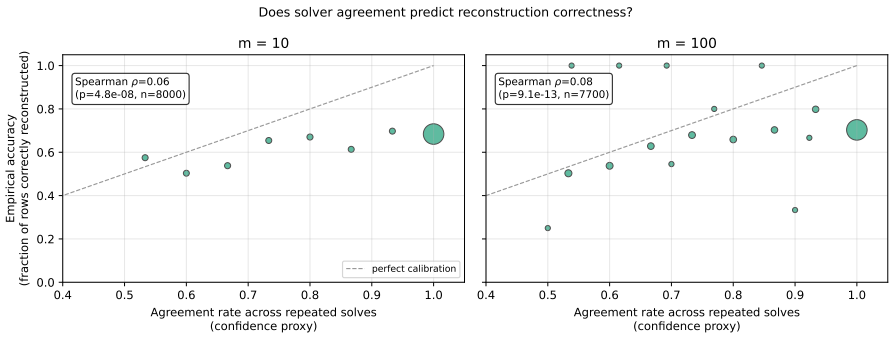


Figure 3: Empirical accuracy vs. agreement rate across 15 repeated solves (confidence proxy), by m. Point size is proportional to bin count. The dashed line is perfect calibration (accuracy = agreement).

With $N_{\text{repeats}} = 15$, agreement rate takes 8 well-populated values per m (roughly 8/15 through 15/15; a handful of additional fractions appear from the rare cell where fewer than 15 repeats solved within the timeout, each with only a few rows – see Limitations). The trend is upward but not strictly monotonic bin-to-bin: at $m = 10$, accuracy moves from 0.57 at agreement 8/15 to a peak of 0.70 at 14/15 before settling at 0.68 in the dominant full-agreement bin (15/15, $n = 6{,}827$); at $m = 100$ it rises from 0.50 at 8/15 to 0.80 at 14/15, then drops back to 0.70 at 15/15 ($n = 5{,}768$) – so even the best-populated bin is not the most accurate one. Row-level Spearman correlation between agreement and correctness is positive and highly significant at both, though modest in magnitude: $\rho = 0.06$ ($p = 4.8 \times 10^{-8}$, $n = 8{,}000$) at $m = 10$, and $\rho = 0.08$ ($p = 9.1 \times 10^{-13}$, $n = 7{,}700$) at $m = 100$.

This is a real, useful confidence signal – an attacker (or an auditor) can trust a unanimous reconstruction meaningfully more than a contested one – but it is not a *calibrated probability*: every curve in Figure 3 sits below

the perfect-calibration line, so a row with full agreement across all 15 solves is correct about 68–70% of the time, not close to 100%. The agreement rate should be read as a valid relative ranking of confidence, not a literal probability of correctness.

2.4 From raw agreement to an actual calibrated probability

The previous subsection established correlation, not calibration, and did not separate the data used to observe the relationship from data used to evaluate it. We close that gap directly: split the 20 seeds into 14 training and 6 held-out test seeds, fit an isotonic regression mapping agreement rate to $P(\text{correct})$ on the training seeds only (separately for $m = 10$ and $m = 100$), and evaluate strictly on the 6 test seeds the mapping never saw.

m	raw Brier	calibrated Brier	raw ECE	calibrated ECE
10	0.320	0.221	0.310	0.016
100	0.331	0.231	0.318	0.072

Table 2: Calibration quality on held-out test seeds: raw agreement rate used directly as a probability, vs. the isotonic-calibrated probability fit on the disjoint training seeds. Lower is better for both metrics.

The calibrated probability roughly halves the Brier score and cuts expected calibration error by a factor of 4–20× (Table 2), evaluated entirely out-of-sample. Figure 4 shows why: the isotonic curve (fit on training seeds) tracks the held-out test points far more closely than the raw agreement value would, which – used directly as a probability – is systematically overconfident across the whole range.

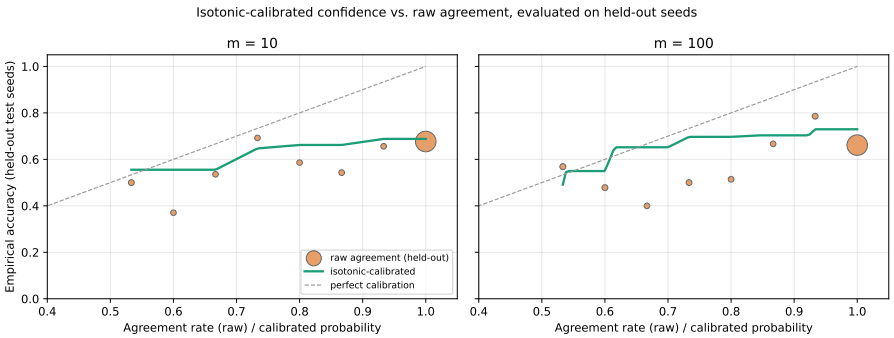


Figure 4: Isotonic-calibrated probability (fit on 14 training seeds) vs. raw agreement rate, both evaluated against empirical accuracy on 6 held-out test seeds never used to fit the calibration.

This is, to our knowledge, the first attempt in this literature to produce

an actual calibrated, validated per-record confidence value for a reconstruction attack against an ensemble of released models (as opposed to either no confidence measure at all, or an uncalibrated single-model entropy measure). We initially ran this with $N_{\text{repeats}} = 7$ (four well-populated agreement levels) and asked whether a finer-grained proxy would improve the calibration; doubling to $N_{\text{repeats}} = 15$ (eight levels, Table 2) left Brier score and ECE essentially unchanged on held-out seeds. Agreement-level resolution was therefore not the bottleneck – the limiting factor is more likely the held-out test-set size (6 seeds), not the coarseness of the confidence scale.

Limitations. (i) The pooled Wilcoxon tests treat each (seed, m) pair as an independent observation; the five m-values drawn from the same seed reuse the same 100-row sample and overlapping trees from the same Rashomon set, so pooled p-values are a convenient summary rather than a fully independent combined test – the per-m, Holm-corrected tests are the safer basis for any individual claim. (ii) At $m = 100$, the *None* condition most often timed out (CP-SAT returned `UNKNOWN` within the 120s budget); cell sizes shrink from 200 at $m = 5$ to 156 at $m = 100$ accordingly, and a handful of repetitions had Rashomon sets smaller than 50/100 trees and were excluded at those m values entirely. (iii) A handful of odd-fraction agreement values (e.g. 0.50, 0.69, 0.85 at $m = 100$) come from the rare cell where fewer than 15 repeats solved within the timeout and have only 1–5 rows each; the reported trend is supported by the eight well-populated $k/15$ bins (hundreds to thousands of rows each), and even among those the relationship is non-monotonic bin-to-bin, not a clean staircase. (iv) The curated 13-column feature set and single enumeration-order tree selection (no diversity-aware strategy) remain as before. (v) The held-out calibration test set is only 6 seeds ($n \approx$ 2,300–2,400 rows); doubling the agreement-level resolution from four to eight levels (reported above) did not improve held-out Brier/ECE, so the test-set size – not the confidence scale's granularity – is the more likely remaining bottleneck before treating the calibrated probabilities as precise.